

# STM32F0 series Entry-level Cortex™-M0 MCU



STM32<sup>®</sup> Releasing your creativity



STMicroelectronics

**STM32F0 MCUs with STM32 DNA<sup>®</sup> at budget price**  
**Now get 32-bit performance at 8/16-bit price**

|                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>System</b><br>Power supply<br>1.8 V regulator<br>POR/PDR/PVD<br>RTC oscillators<br>32 kHz + 4 - 32 MHz<br>Internal RC oscillators<br>40 kHz + 8 MHz<br>PLL<br>Clock control<br>Calendar RTC<br>SysTick timer<br>2x window watchdogs<br>(independent and system)<br>27/39/55 I/Os<br>Touch-sensing controller<br>Cyclic redundancy<br>check (CRC) | <b>Up to 64-Kbyte Flash memory</b><br>Up to 8-Kbyte SRAM<br>HW parity checking<br>20-byte backup data                                                                     |
| <b>Control</b><br>1x 16-bit motor control<br>PWM<br>Synchronized AC timer<br>1x 32-bit timer<br>5x 16-bit PWM timers<br>1x 16-bit basic timer                                                                                                                                                                                                       | <b>ARM Cortex-M0 CPU</b><br>48 MHz<br>Nested vector<br>interrupt<br>controller (NVIC)<br>SW debug<br>ARM-Lite bus matrix<br>APB bus<br>5-channel DMA                      |
|                                                                                                                                                                                                                                                                                                                                                     | <b>Connectivity</b><br>HDMI CEC<br>2x SPI<br>(1x with IFS mode)<br>2x I2C<br>(1x with fast-mode plus)<br>2x USART<br>with modem control<br>(1x with LIN, smartcard, IrDA) |
|                                                                                                                                                                                                                                                                                                                                                     | <b>Analog</b><br>1-channel 12-bit DAC<br>1x 12-bit ADC<br>16 channels / 1 MSPS<br>2x analog comparator                                                                    |

## Key features

Core and operating conditions

ARM® Cortex™-M0

0.9 DMIPS/MHz up to 48 MHz

1.8/2.0 to 3.6 V supply range

High-performance connectivity

6 Mbit/s USART 18 Mbit/s SPI

with 4- to 16-bit data frame

1 Mbit/s I2C fast-mode plus HDMI CEC

Enhanced control

16-bit 3-phase PWM motor control timer

5x 16-bit PWM timers 16-bit basic timer

32-bit PWM timer 12 MHz I/O toggling

## Targeted applications

- Home automation
- Home appliances
- Motor controls
- Sensors
- A/V receivers, TVs
- Remote controls
- Gaming

## STM32F0-Discovery Kit

Everything you need to discover  
STM32F0 32-Bit ARM Cortex-M0  
based MCUs having embedded  
real time ST-Link/V2  
programmer/debugger



Please post your queries to: [mcu-enquiry@st.com](mailto:mcu-enquiry@st.com)

[www.st.com/mcu](http://www.st.com/mcu)